

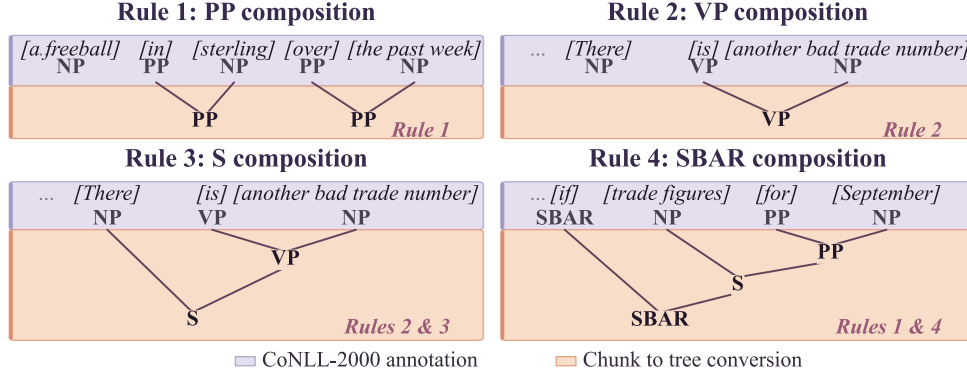


Fig. S1. Chunk-to-tree conversion rules.

APPENDIX A CHUNK-TO-TREE CONVERSION

The CoNLL-2000 dataset [A1] provides gold-standard chunk annotations derived from the Penn Treebank [A2], and is publicly available. However, CoNLL-2000 provides only flat chunk annotations without hierarchical structure. We convert these chunks into shallow constituency trees using four linguistically motivated composition rules (Fig. S1):

- 1) **PP composition.** A PP chunk immediately followed by an NP chunk is merged into a prepositional phrase: (PP (NP)).
- 2) **VP composition.** A VP chunk is merged with the immediately following chunk to form a verb phrase: (VP (XX)).
- 3) **S composition.** An NP chunk followed by a VP chunk is merged into a sentence node: (S (NP) (VP)).
- 4) **SBAR composition.** An SBAR chunk is merged with the following chunk to form a clausal structure: (SBAR (S (XX))).

These rules are applied iteratively to produce shallow constituency trees. Although the resulting trees are shallow, their constituent boundaries are inherited from gold-standard Penn Treebank annotations and are therefore linguistically reliable.

APPENDIX B EXPERIMENTAL DETAILS

Model access. For evaluation with general tests, LLMs (the Qwen family: qwen2.5-flash, qwen2.5-plus, and qwen2.5-max) were accessed through the Alibaba API. For evaluation with diagnostic tests, LLMs (qwen2.5-max, deepseek-v4-pro, gpt-5.5, claude-opus-4.7, gemini-3.1-pro, and grok-4.20-0309) were accessed through the OpenRouter API and the Alibaba API.

Generation parameters. Temperature was set to 0. General tests used a 256-token completion limit; diagnostic tests used the provider default. Top- p and all other parameters used provider defaults.

Trials and runs. Following Liu et al. [A3], for each test we constructed 24 trials per run, where each trial paired a

demonstration with a test sentence following the construction procedure specific to that test. For the general tests, each LLM was evaluated over 600 runs per test (6 prompts $\times$ 100 runs each), yielding 14,400 responses per LLM per test. For the diagnostic tests, each LLM was evaluated over 30 runs per test using a single prompt (i.e., Prompt A), yielding 720 responses per LLM per test. The conversation history was cleared between trials to ensure independence.

APPENDIX C STATISTICAL ANALYSIS AND CHANCE LEVEL

Statistical tests and confidence intervals. For the between-condition comparisons reported in Figs. 3 and 4, we applied two-sided permutation tests, using 10,000 permutations. When observations from the two conditions could be paired one-to-one by trial, we generated the null distribution by randomly flipping the sign of each paired difference. For unpaired observations, we generated the null distribution by permuting the condition labels. For Test 1, we compared the observed constituent rate with a Monte Carlo null distribution generated by uniformly sampling one eligible span per trial (10,000 iterations). The one-sided p -value was the proportion of null rates at least as large as the observed rate, with a plus-one correction. The bar plots show 95% Wilson score confidence intervals. Bootstrap confidence intervals in the processed-result summaries were estimated with 1,000 resamples [A4].

Chance level for single-constituent deletions. For Tests 1, 4, and 5, we calculated the chance level for single-constituent deletions. For each sentence, we enumerated all non-empty contiguous spans shorter than the full sentence and assumed that each eligible span was equally likely to be selected. The chance level was the exact proportion of these eligible spans that matched a complete constituent in the corresponding constituency tree.

Chance level for constituent localization. For Test 3, we calculated the chance level of deleting the corresponding constituent. For each sentence, we enumerated all non-empty contiguous spans shorter than the full sentence and assumed that each eligible span was equally likely to be selected. The chance level was the reciprocal of the number of these eligible spans.

TABLE S1

SENTENCE STATISTICS FOR TESTS 1–6; LENGTHS ARE IN WORDS. TESTS 4–5 POOL LANGUAGES AND TEST 6 POOLS AMBIGUITY CONDITIONS.

Test	Sentence type	#	Mean length	Range
Test 1	Demonstration	176	8.76 $\pm$ 2.79	4–14
Test 1	Test sentence	183	8.85 $\pm$ 2.99	4–14
Test 2	Demonstration	155	6.67 $\pm$ 2.04	3–12
Test 2	Test sentence	54	7.94 $\pm$ 1.71	4–12
Test 3	Demonstration	43	9.12 $\pm$ 2.33	5–14
Test 3	Test sentence	40	9.25 $\pm$ 2.49	4–14

Test	Sentence type	#	Mean length	Range
Test 4	Demonstration	60	6.55 $\pm$ 1.08	5–9
Test 4	Test sentence	60	7.17 $\pm$ 1.06	5–10
Test 5	Demonstration	60	6.55 $\pm$ 1.08	5–9
Test 5	Test sentence	60	7.28 $\pm$ 1.06	5–10
Test 6	Demonstration	60	6.42 $\pm$ 1.48	5–9
Test 6	Test sentence	60	9.82 $\pm$ 2.20	7–15

TABLE S2

PROMPTS USED IN THE WORD DELETION TASK. PROMPTS A–C ARE ADOPTED FROM LIU ET AL. [A3]; PROMPTS D–F ARE NEWLY DESIGNED. EACH PROMPT ENDS WITH “ANSWER DIRECTLY WITHOUT EXPLANATION.”

Variant	Prompt
A	John developed a very special way of speaking. For ‘{demonstration sentence}’, he would say ‘{deleted sentence}’. Please carefully examine John’s speaking style, and guess what he would say for the sentence: ‘{test sentence}’. Answer directly without explanation.
B	A man speaks in a really weird way after a stroke. For ‘{demonstration sentence}’, he would say ‘{deleted sentence}’. What will he say for the sentence: ‘{test sentence}’? Answer directly without explanation.
C	An alien is learning our language, but he tends to omit some words in his sentences. For ‘{demonstration sentence}’, he would say ‘{deleted sentence}’. Please guess how the alien would express the following sentence: ‘{test sentence}’. Answer directly without explanation.
D	Someone shortens English sentences only by omitting words from the original sentence—without substituting new synonyms, adding new ideas, or reordering the whole clause. Example: full sentence: “{demonstration sentence}”. Shortened by omission in the same style: “{deleted sentence}”. Now shorten this sentence in that same way (omit words only): “{test sentence}”. Answer directly without explanation.
E	You follow an editorial checklist: the shorter line must be an erase-only version of the longer one—keep surviving words in the same order, add no new clauses, and do not swap in synonyms that effectively rewrite the predicate. Example: full sentence: “{demonstration sentence}”. Shortened by omission in the same style: “{deleted sentence}”. Now shorten this sentence in that same way (omit words only): “{test sentence}”. Answer directly without explanation.
F	Treat this as tightening prose under a brevity preference: shorten by dropping words from the source only. What remains should read as the same surface wording where it appears—no paraphrase-as-rewrite and no stitched-in new phrases. Example: full sentence: “{demonstration sentence}”. Shortened by omission in the same style: “{deleted sentence}”. Now shorten this sentence in that same way (omit words only): “{test sentence}”. Answer directly without explanation.

TABLE S3

DETAILED TEST 1 RESULTS BY PROMPT. “SINGLE,” “MULTI,” “PARTIAL,” AND “C+P” DENOTE A SINGLE CONSTITUENT, MULTIPLE CONSTITUENTS, A PARTIAL CONSTITUENT, AND A CONSTITUENT PLUS A PARTIAL CONSTITUENT, RESPECTIVELY.

Model	Prompt	Single	Multi.	Partial	C+P	Other
Qwen Max	A	.43	.00	.36	.15	.07
	B	.46	.00	.33	.14	.08
	C	.38	.00	.43	.11	.09
	D	.56	.01	.29	.09	.06
	E	.52	.00	.33	.07	.08
	F	.50	.01	.33	.16	.01
Qwen Plus	A	.52	.00	.27	.06	.14
	B	.62	.01	.28	.04	.05
	C	.44	.00	.36	.04	.16
	D	.37	.00	.22	.05	.36
	E	.34	.01	.23	.05	.36
	F	.47	.00	.30	.08	.14
Qwen Flash	A	.21	.01	.14	.04	.60
	B	.17	.01	.12	.03	.67
	C	.40	.01	.31	.08	.20
	D	.27	.00	.31	.06	.35
	E	.48	.00	.27	.04	.20
	F	.17	.00	.35	.04	.43

TABLE S4

DETAILED TEST 1 RESULTS FOR QWEN MAX BY NUMBER OF DEMONSTRATIONS. ABBREVIATIONS FOLLOW THE PRECEDING TABLE.

#	Demo	Prompt	Single	Multi.	Partial	C+P	Other
1	A		.43	.00	.36	.15	.07
1	B		.46	.00	.33	.14	.08
1	C		.38	.00	.43	.11	.09
1	D		.56	.01	.29	.09	.06
1	E		.52	.00	.33	.07	.08
1	F		.50	.01	.33	.16	.01
2	A		.55	.03	.22	.19	.00
2	B		.61	.01	.21	.17	.00
2	C		.56	.03	.26	.14	.00
2	D		.61	.01	.28	.08	.02
2	E		.61	.00	.29	.06	.03
2	F		.63	.01	.26	.09	.01
5	A		.68	.03	.15	.14	.00
5	B		.71	.01	.17	.11	.00
5	C		.67	.02	.18	.13	.00
5	D		.73	.01	.20	.06	.01
5	E		.69	.01	.24	.06	.00
5	F		.65	.01	.22	.08	.04
10	A		.71	.02	.14	.12	.00
10	B		.73	.02	.14	.10	.00
10	C		.69	.03	.15	.13	.00
10	D		.76	.01	.15	.08	.01
10	E		.71	.01	.22	.06	.01
10	F		.70	.02	.19	.09	.00

TABLE S5

DETAILED RESULTS OF TESTS 2 AND 3 BY PROMPT. TEST 2 REPORTS THE EXPLAINED RATIO AS PARENT-CATEGORY RULE / NODE-CATEGORY RULE.

Model	Prompt	Test 2	Test 3
Qwen Max	A	.55 / .09	.33
	B	.61 / .10	.39
	C	.51 / .11	.30
	D	.36 / .18	.53
	E	.19 / .23	.44
	F	.42 / .22	.40
Qwen Plus	A	.50 / .15	.41
	B	.55 / .12	.42
	C	.43 / .13	.25
	D	.22 / .12	.36
	E	.25 / .13	.35
	F	.32 / .26	.38
Qwen Flash	A	.09 / .11	.30
	B	.09 / .08	.26
	C	.24 / .17	.40
	D	.19 / .13	.27
	E	.34 / .16	.40
	F	.14 / .10	.20

TABLE S7

DETAILED RESULTS OF TEST 6. THE PROPORTION OF TARGET-STRING DELETIONS.

Ambiguity	Model	Structure 1	Structure 2
PP attachment	Qwen Max	.33	.40
	DeepSeek V4	.91	.76
	Claude 4.7	.42	.32
	GPT-5.5	.93	.75
	Gemini 3.1	.96	.81
	Grok 4.20	.88	.79
Adjunct attachment	Qwen Max	.03	.16
	DeepSeek V4	.93	.95
	Claude 4.7	.40	.59
	GPT-5.5	.88	.91
	Gemini 3.1	.85	.88
	Grok 4.20	.82	.86

TABLE S6

DETAILED RESULTS OF TESTS 4 AND 5. EXPLAINED RATIOS ARE REPORTED AS PARENT-CATEGORY RULE / NODE-CATEGORY RULE.

Lang.	Model	Constituent rate		Explained ratio	
		T4	T5	T4	T5
Chinese	Qwen Max	.57	.45	.13/.82	.07/.58
	DeepSeek V4	.96	.93	.38/.59	.27/.68
	Claude 4.7	.80	.81	.16/.79	.24/.65
	GPT-5.5	.86	.75	.22/.72	.20/.68
	Gemini 3.1	.94	.85	.31/.65	.21/.70
	Grok 4.20	.91	.87	.25/.73	.21/.75
English	Qwen Max	.96	.90	.96/.04	.93/.07
	DeepSeek V4	.98	.95	.97/.03	.90/.08
	Claude 4.7	.94	.68	.84/.12	.58/.12
	GPT-5.5	.75	.62	.82/.12	.70/.17
	Gemini 3.1	.82	.57	.83/.10	.69/.16
	Grok 4.20	.77	.63	.89/.07	.81/.11

APPENDIX REFERENCES

- [A1] E. F. Tjong Kim Sang and S. Buchholz, "Introduction to the CoNLL2000 shared task chunking," in *Fourth Conference on Computational Natural Language Learning and the Second Learning Language in Logic Workshop*, 2000.
- [A2] M. P. Marcus, B. Santorini, and M. A. Marcinkiewicz, "Building a large annotated corpus of English: The Penn Treebank," *Computational Linguistics*, vol. 19, no. 2, pp. 313–330, 1993.
- [A3] W. Liu, M. Xiang, and N. Ding, "Active use of latent tree-structured sentence representation in humans and large language models," *Nature Human Behaviour*, vol. 10, no. 2, pp. 303–316, Feb 2026.
- [A4] B. Efron, "Better bootstrap confidence intervals," *Journal of the American Statistical Association*, vol. 82, no. 397, pp. 171–185, 1987.